

Smart Parking Platform Project - Software corp.

Homework 1

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1. Existing Software Landscape Research

SpotHero

SpotHero is a parking software that lets users search and compare parking for places like airports, events, and cities. What's special about this software is that users are able to reserve the parking ahead of time and they will receive a parking pass for the reserved area.

The software does lack a feature where it tells drivers whether a parking area is almost full or empty. Our Smart Parking Platform will be able to let users see the current number of available spaces a parking area has. We'll have the platform update periodically in order to give the user the latest information.

ParkMobile

ParkMobile is also a platform where it lets users pay for parking with their mobile devices and allows them to extend their reservation parking time. It also provides reports about users, what vehicle they want to reserve with, and expenses.

One problem is that their parking information is too broad for the user, it only provides the location of the parking and the cost. We can improve this by adding a parking rating system, so users can see which parking area is recommended. These ratings and reviews are going to be produced by the user to have a better outcome.

ParkWhiz

Drivers are able to compare choices and pay by scanning a QR code or paying with their phones ahead of time. The app also shows garage instructions and available features.

One problem of the app is that there are very limited places where you can park ahead of time, most of the places usually required by scanning a QR code while you are at the parking or offering a “Park then Pay” option. We will make sure that all parking areas have an option to reserve and pay ahead of time, rather than having a selected few.

2. Vision and Scope

2.1 Vision

Software corp. aim to help drivers make parking decisions instead of giving them a hard time on where to park. Our goal is to give the drivers the ability to compare parking facilities, see current availability information, see parking recommendations, and pay for parking ahead of time.

2.2 Scope

- Register license plates, vehicles, and provide authentication.
 - Provide an interactive map of prices, availability, and reviews of every parking area.
 - Include payment feature for digital payments, provide confirmation, reservation history, and cancellation feature.
 - Update the information frequently to give drivers the latest information
 - A system where we can track how many vehicles are in an area.
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3. User Cases

User Case 1: Creating an Account

- **Actor:** Driver/User
- **Goal:** Create an account in order to use it.

- **Precondition:** The driver needs to have access to the platform.
- **Steps/Trigger:** The driver has to enter their information, enable their location permission.
- **Post Conditions:** The driver has successfully created an account.

User Case 2: Logging In

- **Actor:** Driver
- **Goal:** Getting into the platform with their account.
- **Precondition:** The driver needs to have an account registered.
- **Steps/Trigger:** The driver will enter their email/username and password.
- **Post Conditions:** The driver is logged in to the platform.

User Case 3:

- **Actor:** Driver
- **Goal:** See how many spaces are available in a parking area.
- **Precondition:** The driver needs access to the map and parking options.
- **Steps/Trigger:** The driver will have to select a parking facility.
- **Post Conditions:** The platform will show the current availability

User Case 4: Looking for Parking

- **Actor:** Driver
- **Goal:** Finding a parking near their destination
- **Precondition:** They need to allow location permission for the platform to access the map.
- **Steps/Trigger:** The driver will need to navigate their map and include what time they want to reserve a parking spot. They will choose which parking is suitable.
- **Post Conditions:** The platform will show the information of the parking spot that the driver has chosen.

User Case 5: Compare Parking Options

- **Actor:** Driver
- **Goal:** Choose the most suitable parking.
- **Precondition:** Access to the maps on the platform.
- **Steps/Trigger:** The Driver will need to choose a parking area.
- **Post Conditions:** The platform will display reviews from other drivers regarding the parking spot.

User Case 6: Reserve Parking

- **Actor:** Driver
- **Goal:** Reserve a parking spot for a certain time.
- **Precondition:** The driver selects the parking area.
- **Steps/Trigger:** The user will select a time slot of when they are going to park.
- **Post Conditions:** The parking space is reserved.

User Case 7: Paying for parking

- **Actor:** Driver
- **Goal:** Pay for the reservation.
- **Precondition:** They have reserved a parking spot.
- **Steps/Trigger:** The Driver will have to enter payment information, then confirm their payment.
- **Post Conditions:** A confirmation will show up that they have paid for the parking.

User Case 8: Change Reservation

- **Actor:** Driver
- **Goal:** Changing reservation time.
- **Precondition:** If a reservation is still active.
- **Steps/Trigger:** Driver will select their reservation, then enter their new time slot.
- **Post Conditions:** Confirmation will pop up that they have updated their reservation.

User Case 9: Cancel a Reservation

- **Actor:** Driver
- **Goal:** Cancel any upcoming reservation they have.
- **Precondition:** The driver has a reservation.
- **Steps/Trigger:** The driver will select the reservation, then press “Cancel Reservation.”
- **Post Conditions:** A confirmation will show up that they have canceled their reservation.

User Case 10: Viewing Parking History

- **Actor:** Driver
- **Goal:** Review previous reservations.
- **Precondition:** The driver has made a reservation in the past.
- **Steps/Trigger:** The driver will go to their Reservation History tab.
- **Post Conditions:** A list of past reservations will display.

User Case 11: Notifications

- **Actor:** Driver
- **Goal:** Receiving updates about a reservation
- **Precondition:** Enable notifications on their device.
- **Steps/Trigger:** Go to settings and turn on notifications.
- **Post Conditions:** The driver will receive notifications about the reservation.

User Case 12: Direction of Reservation

- **Actor:** Driver
- **Goal:** Navigating to the destination.
- **Precondition:** The Driver selected a parking facility.
- **Steps/Trigger:** The Driver will press on the direction button.
- **Post Conditions:** The map opens with the direction to the parking area.

User Case 13: Update Parking Information

- **Actor:** Operator
- **Goal:** Updating parking information to have the accurate information.
- **Precondition:** The operator has permissions to update information.
- **Steps/Trigger:** The operator updates the capacity and availability.
- **Post Conditions:** New information is saved and displayed for the drivers.

User Case 14: Parking Prices

- **Actor:** Operator
- **Goal:** Updating parking prices.
- **Precondition:** The operator has permission to update the prices.
- **Steps/Trigger:** The operator enters the new price, then confirms it.
- **Post Conditions:** The new price is saved and displayed for the drivers.

User Case 15: View Information Reports

- **Actor:** Operator
- **Goal:** Review parking activity and overall performance.
- **Precondition:** The operator has access to the information.
- **Steps/Trigger:** The operator selects an area.
- **Post Conditions:** The platform will showcase the performance and timeline of the selected parking space.